

WASP-52b

R-0522

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230730 · created 2026-07-21T00:33:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460155.8974 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.062
Transit depth (Rp/Rs)^2	4.64 +/- 2.68 [%]
Orbital Inclination (inc)	86.87 +/- 2.68
Airmass coefficient 1 (a1)	1.061 +/- 0.067
Airmass coefficient 2 (a2)	-0.04 +/- 0.14
Scatter in the residuals of the lightcurve fit is	6.41 %
Best Comparison Star	#1 - [419, 293]
Optimal Aperture	6.37
Optimal Annulus	6.63
Transit Duration (day)	0.083 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.062 vs 0.1646 (deviation 0.81σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.48σ)
Mid-time O–C	16.5 min
Depth SNR	3.5
Residual scatter	6.41 %

Light curve

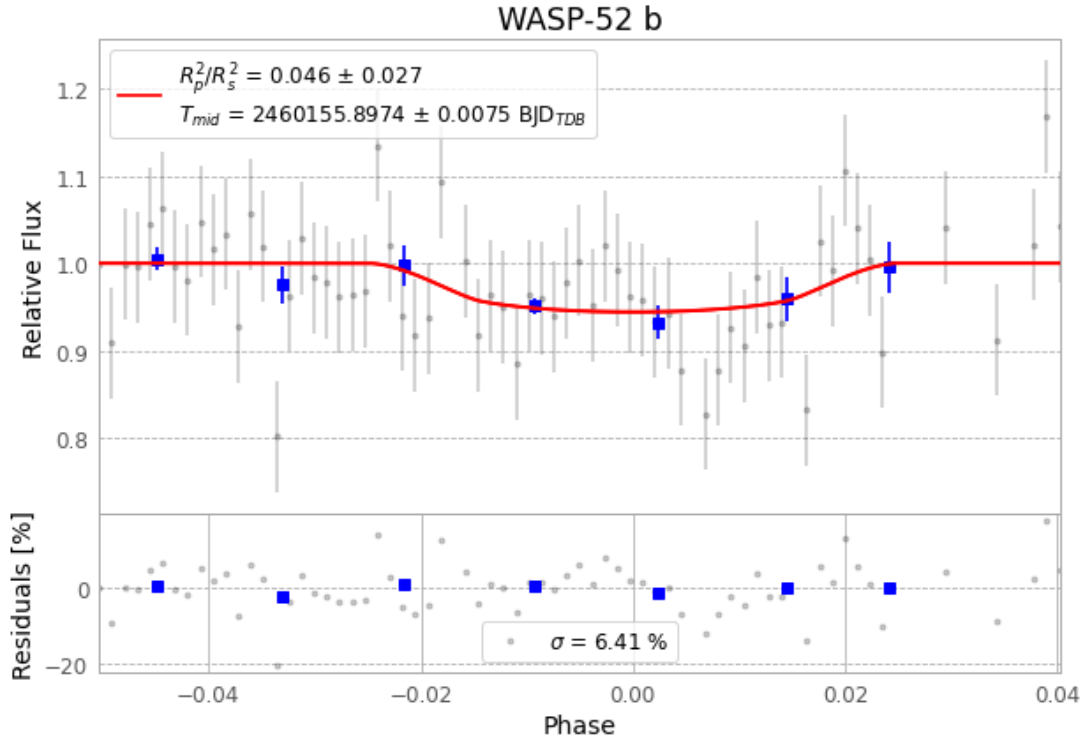


Figure 1 — FinalLightCurve_WASP-52 b_2023-07-30.png (SHA-256 8f5aef7f86ece561...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 211], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4fb2f4678c498232...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 4685edf

Data provenance

77 FITS frames (51 MB), WASP-52230730071815.FITS → WASP-52230730110616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230730.log (67ea35c09a4e...), FinalLightCurve_WASP-52 b_2023-07-30.png (8f5aef7f86ec...), FinalLightCurve_WASP-52 b_2023-07-30.csv (ba230efa4fd4...)

Compute resources

Wall time 330.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 00:33Z.